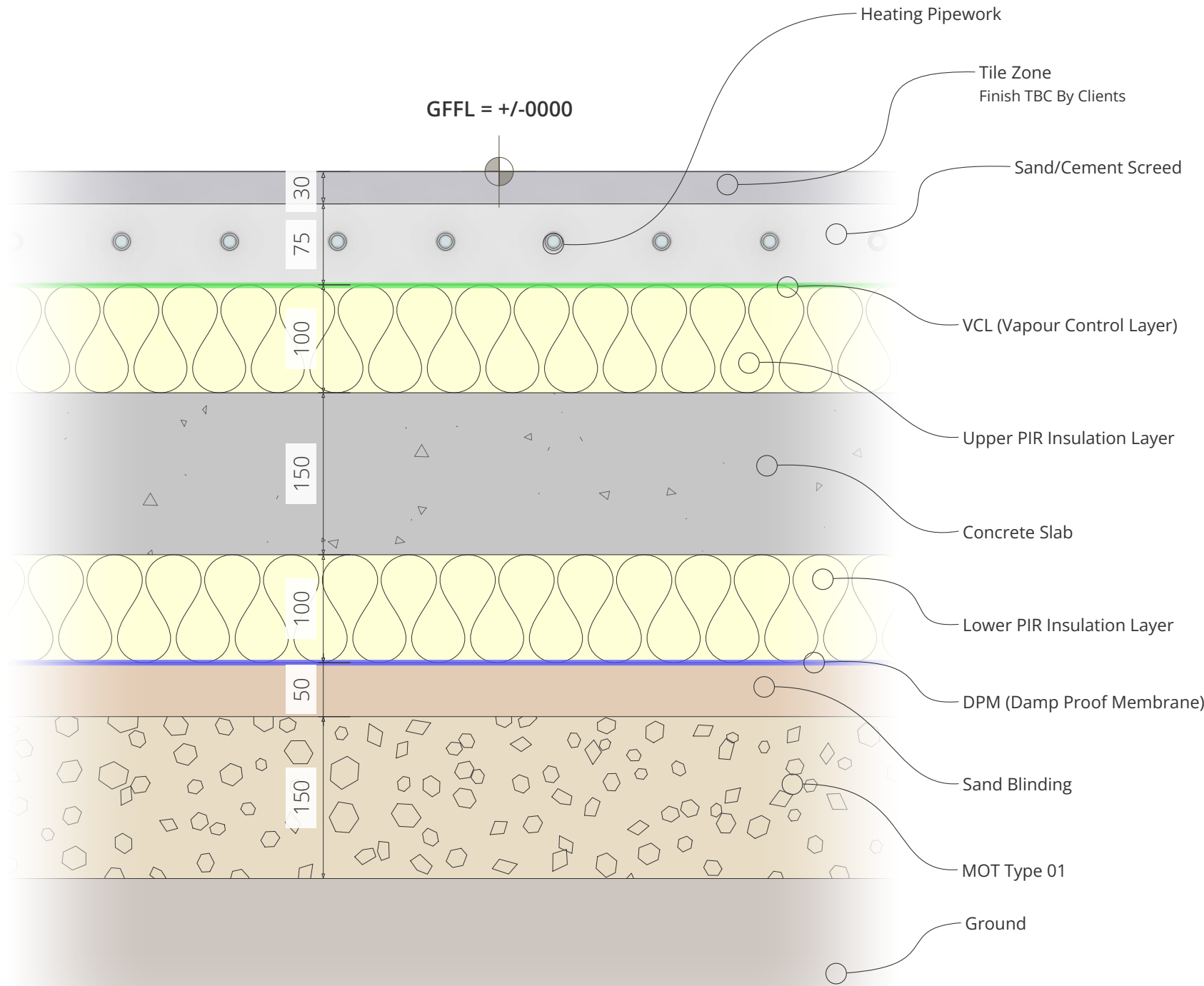




GF10 - Insulated & Heated Floor Slab

Calculation			
U = 1 / (0.17 + 0.023 + 0.065 + 4.545 + 0.111 + 4.545 + 0.040) = 1 / 9.50 = 0.105 W/m²K			
Layer	Thickness (mm)	Lambda (W/mK)	R-value (m²K/W)
Internal Surface Resistance (Rsi) - Downward Flow	-	-	0.170
Floor Finishes - Tile/Adhesive Allowance	30	1.30	0.023
Screed - Sand and Cement (1:4 Mix)	75	1.15	0.065
Separation Layer - VCL (Polyethylene)	0	0.33	0.000
Upper Thermal Insulation - PIR (Celotex GA4000)	100	0.022	4.545
Structural Concrete Slab - Cast-in-situ (GEN1)	150	1.35	0.111
Lower Thermal Insulation - PIR (Celotex GA4000)	100	0.022	4.545
Damp Proof Membrane - 1200g	0	0.33	0.000
Blinding - Sand	50	-/-	-/-
Hardcore Sub-base - MOT Type 1	150	-/-	-/-
External Surface Resistance (Rse)	-	-	0.040
Total R-value			9.499
Result			
U-Value			0.105 W/m²K

✓

Value Compliant - England Part L1B : Wall U-value 0.18 W/m²K.

Element	Specifications
Floor Finish	Final floor finish to Client's choice (e.g. Tiled, brick tiles). 30mm Allowance. All finishes, adhesives and grouts must be flexible and certified compatible with Underfloor Heating (UFH). CRITICAL: Prior to tiling, install a high-performance uncoupling/decoupling membrane (e.g. Schluter-DITRA 25) directly over the primed screed. This is essential to prevent cracking and to accommodate UFH thermal movement.
Sand/Cement Screed	75mm traditional fibre-reinforced sand and cement screed. Mix ratio 1:3 to 1:4 (Cement:Sharp Sand) incorporating polypropylene fibres (PP) to minimize shrinkage cracking. Surface finish to be wood float or power float to BS 8204-1 class SR2 tolerance (±5mm under 3m straight edge). Include movement joints at door thresholds and where bay sizes exceed 40m² or 8m linear length. This layer acts as the primary thermal emitter for the UFH system.
Underfloor Heating & Perimeter Strip	Hydronic Underfloor Heating pipework clipped to the upper insulation layer. CRITICAL: Install a continuous 25mm PIR or polyethylene perimeter expansion strip around all external and internal wall abutments. This strip must turn up the full depth of the screed against the 100mm inner blockwork leaf to isolate the heated screed from the masonry.
Separation Layer (VCL)	500 gauge (125 micron) polythene Vapour Control Layer (VCL). Lay directly over the upper insulation boards with 150mm taped overlaps. Turn up at perimeters to the height of the skirting. This layer protects the foil face of the insulation from alkaline screed attack and prevents wet screed migration between board joints.
Upper Thermal Insulation	100mm High-Performance PIR Insulation such as Celotex GA4000, or similar rigid polyisocyanurate). Thermal conductivity 0.022 W/mK. Install tightly butted directly on top of the concrete slab. This specific layer is positioned to reflect UFH heat upwards immediately, ensuring rapid system response times.
Structural Concrete Slab	150mm Cast-in-situ Concrete Slab. Concrete Mix GEN1 or ST2 to BS 8500-1. Finish surface with a tamper or skip float to provide a level base for the upper insulation. This slab forms the structural deck, thermally isolated from the ground by the lower insulation layer.
Lower Thermal Insulation	100mm Celotex GA4000 (or equivalent General Purpose PIR). Install tightly butted directly over the DPM. Taping all joints with foil tape is mandatory to prevent concrete grout bleeding between boards. Important workmanship note: Unlike XPS, this board is susceptible to point-load damage and moisture intake if compromised. The Contractor MUST inspect the entire surface for punctures, dents, or crushed edges immediately prior to the concrete pour. Any damage to the foil facing or foam core must be repaired or the board replaced. Use temporary walking boards for wheelbarrows and foot traffic during the pour to prevent crushing the insulation matrix. Ensure the DPM below is absolutely pristine; if the DPM fails, this insulation can absorb groundwater over time.
Damp-Proof Membrane	1200 gauge (300 micron) heavy-duty polythene DPM (e.g., Visqueen). Lay on the sand blinding. Sheets to be jointed with double-sided bitumen tape and single-sided overlap tape (minimum 150mm overlap). The DPM must turn up the internal face of the 100mm inner blockwork skin a minimum of 150mm to lap seamlessly with the horizontal DPC specified in the WT10 Wall Spec. Ensure corner folds are neat and sealed to maintain gas/moisture integrity.
Blinding	50mm Sand Blinding. Compacted sand layer spread over the hardcore to fill voids and provide a soft, even bed that protects the DPM from puncture by sharp aggregate.
Hardcore Sub-base	150mm minimum MOT Type 1 Granular Sub-base. Lay in layers not exceeding 150mm and mechanically compact to refusal using a vibrating plate. Material to be clean, crushed concrete or stone, certified non-frost susceptible.
Design Rationale	We have consciously upgraded the floor specification to vastly exceed Part L requirements by utilising 200mm of insulation split into two distinct zones. The Upper 100mm layer is positioned to maximise Underfloor Heating efficiency by minimizing downward heat loss into the slab. The Lower 100mm layer thermally decouples the heavy structural concrete slab from the cold ground. This 'belt and braces' approach ensures the structural mass remains warmer than standard ground-bearing slabs while the screed provides a responsive heating surface.
DPM Continuity	A common failure point is the connection between the floor DPM and Wall DPC. Ensure the blockwork inner leaf is cleaned of mortar snots at the base to allow the DPM to turn up flush against the blockwork.
Screed Drying	Traditional 75mm screed takes approx. 75-80 days to dry fully (1mm/day for first 40mm , 0.5mm/day thereafter). Do not lay moisture-sensitive flooring until a moisture test confirms <75% RH.

Building Control Approval Submission

This drawing is submitted for Building Control approval. Construction must not commence until the Inspector sanctions the proposed structural details, thermal performance, and moisture integrity. The Principal Contractor is fully responsible for coordinating all statutory inspections to verify strict compliance with the Building Regulations 2010.